Olympiad Test Paper — Science (Class 7)

Chapter 11: Transportation in Animals and Plants

Subject	Science (NCERT Class 7)
Chapter	11 — Transportation in Animals and Plants
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. All questions stay within the Class 7 syllabus. No calculator is needed — any arithmetic (pulse counts, rates) is designed to work by hand. Most questions need two to four reasoning steps; watch for traps about arteries vs veins, xylem vs phloem, and common misconceptions such as "blood is blue." Do not write on this paper — mark answers on your answer sheet.

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1. The liquid part of blood, in which the cells float, is called:

(A) haemoglobin (B) plasma
(C) serum-free urea (D) lymph cells

2. Blood looks red because:

(A) plasma is red in colour
(B) white blood cells turn red when they fight germs
(C) red blood cells contain haemoglobin, an iron-rich red pigment
(D) the heart muscle stains the blood red

3. Which blood cells defend the body against germs and infection?

(A) red blood cells (B) platelets
(C) white blood cells (D) plasma proteins

4. When you cut your finger, the blood soon stops oozing and forms a clot. The blood component most responsible for this is the:

(A) platelets (B) red blood cells
(C) white blood cells (D) plasma alone

5. The main job of red blood cells is to:

(A) carry oxygen using haemoglobin (B) fight bacteria and viruses
(C) help the blood to clot (D) digest food in the blood

6. A person who is short of iron in the diet may make too little haemoglobin. The most direct effect of this would be:

(A) the blood cannot clot at all (B) the heart stops beating

(C) the white blood cells stop working (D) the blood carries less oxygen to the body

7. Which list correctly matches the components of blood?

(A) RBCs fight germs; WBCs carry oxygen; platelets clot blood

(B) plasma carries oxygen; RBCs clot blood; WBCs are the liquid part

(C) RBCs carry oxygen; WBCs fight germs; platelets help clotting

(D) platelets carry oxygen; plasma fights germs; RBCs clot blood

8. The human heart is divided into:

(A) two chambers (B) three chambers

(C) six chambers (D) four chambers

9. The two upper chambers of the heart are called the:

(A) atria (B) ventricles

(C) valves (D) arteries

10. The familiar "lub-dub" sound of a heartbeat is produced by:

(A) blood rushing through capillaries (B) the pulse felt at the wrist

(C) the lungs filling with air (D) the closing of the heart's valves

11. Why does the heart keep oxygen-rich and oxygen-poor blood on separate sides instead of letting them mix?

(A) so the blood can change colour faster

(B) because mixed blood would be too thick to pump

(C) to make the heartbeat louder

(D) so that fully oxygenated blood can be delivered efficiently to the body

12. The heart is mainly made of:

(A) bone (B) muscle that contracts rhythmically

(C) cartilage (D) fat that stores energy

13. A vessel that carries blood AWAY from the heart is a/an:

(A) vein (B) ureter

(C) capillary (D) artery

14. Arteries have thick, elastic, muscular walls mainly because they:

(A) carry blood very slowly

(B) must withstand blood pumped out at high pressure

(C) always carry deoxygenated blood (D) contain valves along their length

15. Which feature is found in veins but NOT in arteries?

(A) valves that stop blood flowing backward(B) thick muscular walls

(C) a one-cell-thin wall (D) the ability to carry blood at high pressure

16. Exchange of oxygen, food and wastes between blood and body cells happens mainly in the:

(A) arteries (B) veins

(C) capillaries (D) heart chambers

17. Capillaries are ideally suited for exchange because their walls are:

(A) only one cell thick (B) thick and muscular

(C) lined with valves (D) made of bone

18. A student says, "Veins look blue through the skin, so the blood inside them is blue." The correct statement is:

(A) blood is blue inside veins and turns red only in air

(B) only children have blue blood

(C) blood is always red; veins merely look bluish through the skin

(D) blood is green in veins and red in arteries

19. Which statement about a vein is correct?

(A) it always carries oxygen-rich blood at high pressure

(B) it is only one cell thick (C) it carries blood away from the heart

(D) it carries blood back toward the heart and has valves

20. A doctor feels a patient's "pulse" at the wrist. The pulse is actually caused by:

(A) blood draining back through the veins(B) air moving in the lungs

(C) the valves of veins snapping shut

(D) the rhythmic expansion of an artery each time the heart pumps

21. The resting pulse rate of a healthy adult is closest to:

(A) 12 beats per minute (B) 220 beats per minute

(C) 72 beats per minute (D) 7 beats per minute

22. Because each pulse corresponds to one heartbeat, a person whose heart beats 72 times a minute will, in 5 minutes at rest, have a pulse count of about:

(A) 72 (B) 144

(C) 720 (D) 360

23. After running hard, Meera's heart beats faster than at rest. This is because the body now needs:

(A) more oxygen delivered quickly to the muscles

(B) less oxygen, so blood is pumped slower(C) to stop the blood from clotting

(D) to cool the heart down

24. A nurse counts 18 pulse beats in 15 seconds. The pulse rate per minute is:

(A) 72 (B) 33
(C) 18 (D) 90

25. If a doctor measures a resting pulse of 120 beats per minute in a calm patient, this is best described as:

(A) exactly normal for an adult at rest (B) lower than the usual resting rate
(C) higher than the usual resting rate (D) impossible for a living person

26. Excretion is the process of:

(A) taking in food (B) carrying oxygen to the cells
(C) pumping blood around the body
(D) removing harmful waste products from the body

27. The two bean-shaped organs that filter wastes from the blood are the:

(A) kidneys (B) lungs
(C) ventricles (D) capillaries

28. The microscopic filtering units inside a kidney are called:

(A) stomata (B) nephrons
(C) platelets (D) ureters

29. The chief nitrogenous waste removed from the blood by the kidneys and passed out in urine is:

(A) glucose (B) oxygen
(C) urea (D) haemoglobin

30. Put the parts of the urinary system in the order that urine travels through them:

(A) bladder → kidney → ureter → urethra(B) ureter → kidney → urethra → bladder
(C) urethra → bladder → ureter → kidney(D) kidney → ureter → bladder → urethra

31. The tube that carries urine from a kidney down to the urinary bladder is the:

(A) ureter (B) urethra
(C) artery (D) phloem

32. On a hot day you sweat. Apart from cooling the body, sweating also helps to:

(A) remove some water and salts (a form of excretion)
(B) absorb oxygen through the skin (C) clot the blood faster
(D) pump blood to the heart

33. When both kidneys fail, a patient's blood can be cleaned by a machine. This treatment is called:

(A) transpiration (B) circulation
(C) transfusion (D) dialysis

34. Which sequence correctly traces what happens to a waste like urea?

(A) made in body cells → carried by blood → filtered by kidneys → leaves as urine

(B) made in the kidneys → carried by xylem → leaves through stomata

(C) made in urine → carried by ureters → stored in the heart

(D) made in the bladder → carried by arteries → leaves through the lungs

35. Why must the body get rid of urea instead of storing it?

(A) a build-up of urea would be harmful (poisonous) to the body

(B) urea is a useful food the body saves for later

(C) urea is needed to make haemoglobin(D) urea keeps the blood red

36. In a plant, water and dissolved minerals are absorbed from the soil mainly by the:

(A) leaves (B) root hairs

(C) flowers (D) phloem tubes

37. Root hairs are good at absorbing water because they:

(A) are green and make food (B) pump water using muscles

(C) greatly increase the surface area in contact with soil water

(D) carry food down to the roots

38. The plant tissue that carries water and minerals from the roots up to the leaves is the:

(A) phloem (B) stomata

(C) xylem (D) cambium ring

39. Water generally moves through the xylem in which direction?

(A) upward, from roots to leaves (B) only downward, from leaves to roots

(C) sideways only, around the stem (D) in no particular direction

40. Loss of water vapour from the leaves of a plant, mainly through tiny pores, is called:

(A) respiration (B) transpiration

(C) excretion (D) germination

41. The tiny pores on a leaf through which water vapour escapes are the:

(A) root hairs (B) capillaries

(C) nephrons (D) stomata

42. As water evaporates from the leaves, it creates a suction that pulls more water up the stem. This force is called the:

(A) heartbeat (B) transpiration pull

(C) pulse pressure (D) clotting force

43. On a hot, dry, windy day, transpiration from a plant's leaves will generally:

(A) stop completely (B) increase, pulling water up faster

(C) reverse, pushing water down (D) have no effect on water movement

44. The tissue that carries food made in the leaves to all other parts of the plant is the:

(A) phloem
(B) xylem
(C) root hair
(D) stoma

45. Food can travel through the phloem:

(A) only upward to the topmost leaves
(B) only downward to the roots
(C) it does not move at all
(D) both upward and downward, to wherever it is needed

46. Sugar made in a leaf needs to reach a growing root tip and a developing fruit at the same time. The tissue that makes this possible is the:

(A) xylem, because it flows in all directions(B) the stomata
(C) the root hairs
(D) phloem, which can transport food in more than one direction

47. A ring of bark (containing the phloem) is removed all the way around a tree trunk. The part of the tree below the ring is most likely to suffer because:

(A) water can no longer reach the leaves
(B) food made in the leaves can no longer travel down to it
(C) the xylem has been cut
(D) the roots can no longer breathe

48. Which pairing of tissue and cargo is correct?

(A) xylem carries water and minerals; phloem carries food
(B) xylem carries food; phloem carries water
(C) both xylem and phloem carry only water
(D) xylem carries oxygen; phloem carries urea

49. A plant with most of its root hairs damaged will most directly struggle to:

(A) make food in its leaves
(B) absorb water and minerals from the soil
(C) lose water through its leaves
(D) bend toward light

50. Which is the best overall analogy between an animal and a plant transport system?

(A) xylem (water up) is like a one-way delivery pipe; the heart has no plant equivalent of pumping food
(B) xylem is like the veins carrying blood back, phloem is like arteries
(C) phloem is like the kidneys filtering waste(D) root hairs are like the heart's ventricles

51. Which statement about arteries and veins is FULLY correct?

(A) arteries carry blood to the heart; veins carry blood away
(B) arteries carry blood away from the heart, usually oxygenated; veins carry blood back, usually deoxygenated
(C) both arteries and veins carry only deoxygenated blood
(D) arteries have valves; veins do not

52. A student claims, "Veins only carry dirty (waste) blood." The most accurate correction is:

(A) the claim is fully correct

(B) veins carry blood away from the heart at high pressure

(C) veins carry only food, never blood

(D) veins generally carry deoxygenated blood, but the vein from the lungs to the heart carries oxygen-rich blood

53. Compared with an artery, a vein generally has:

(A) the job of carrying blood away from the heart

(B) thicker walls, higher pressure, and no valves

(C) walls only one cell thick (D) thinner walls, lower pressure, and valves

54. Why does a single-celled organism like an Amoeba NOT need a heart, blood vessels or xylem?

(A) it is too small to have organs (B) it does not need oxygen or food

(C) materials can be exchanged directly across its surface by diffusion

(D) it carries blood inside one giant cell

55. A multicellular plant or animal needs a transport system mainly because:

(A) it has no surface for exchange

(B) its inner cells are too far from the surface to get supplies by simple diffusion alone

(C) it never needs oxygen (D) it only lives in water

56. Which row correctly matches each tube with what flows through it?

(A) ureter → blood; xylem → food; vein → urine

(B) ureter → urine; xylem → water; phloem → food

(C) ureter → food; xylem → urine; phloem → water

(D) artery → urine; vein → water; phloem → blood

57. Both the kidneys (in animals) and transpiration (in plants) involve the movement of:

(A) haemoglobin (B) bone

(C) water (D) platelets

58. Blood leaving the heart for the body and blood returning to the heart differ mainly in that the returning blood:

(A) carries more oxygen

(B) carries less oxygen, having given it to the body cells

(C) is no longer red at all (D) has turned into urine

59. A gardener notices a potted plant wilting on a hot afternoon even though the soil is moist. The best explanation is that:

(A) the plant has stopped making food forever(B) the xylem has turned into phloem

(C) the leaves are losing water by transpiration faster than the roots can supply it

(D) the soil water has become poisonous

60. Which overall summary of transport is correct?

(A) in animals veins carry blood away from the heart; in plants phloem carries water up

(B) in animals xylem pumps blood; in plants the heart carries water

(C) in animals the heart pumps blood through vessels; in plants xylem carries water up and phloem carries food

(D) both animals and plants use a heart to move all materials

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End of question paper. Maximum marks: 240. Marking: +4 correct, −1 wrong, 0 unattempted.

Solutions — Science (Class 7), Chapter 11: Transportation in Animals and Plants

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	D	21	C	31	A	41	D	51	B
2	C	12	B	22	D	32	A	42	B	52	D
3	C	13	D	23	A	33	D	43	B	53	D
4	A	14	B	24	A	34	A	44	A	54	C
5	A	15	A	25	C	35	A	45	D	55	B
6	D	16	C	26	D	36	B	46	D	56	B
7	C	17	A	27	A	37	C	47	B	57	C
8	D	18	C	28	B	38	C	48	A	58	B
9	A	19	D	29	C	39	A	49	B	59	C
10	D	20	D	30	D	40	B	50	A	60	C

Answer-key distribution: A = 16, B = 14, C = 14, D = 16.

Worked Solutions

1. (B) Blood is a fluid called plasma in which the cells (RBCs, WBCs, platelets) float. Haemoglobin is a pigment inside RBCs, not the liquid; lymph and "serum-free urea" are distractors. The liquid part is plasma.

2. (C) Red blood cells contain haemoglobin, an iron-rich red pigment that carries oxygen and gives blood its red colour. Plasma is pale yellow; WBCs do not turn red; the heart does not stain blood.

3. (C) White blood cells (WBCs) are the body's defenders against germs and infection. RBCs carry oxygen, platelets help clotting, and plasma is the liquid carrier.

4. (A) Platelets are the blood component that helps the blood to clot and seal a cut. RBCs carry oxygen, WBCs fight germs, and plasma alone cannot form the clot.

5. (A) The main job of red blood cells is to carry oxygen, using the haemoglobin they contain. Fighting germs is WBCs' job; clotting is platelets'; blood does not digest food.

6. (D) Haemoglobin carries oxygen, so making too little of it means the blood carries less oxygen to the body (this is what happens in anaemia). Clotting (platelets) and WBC defence are not directly affected.

7. (C) The correct matching is: RBCs carry oxygen, WBCs fight germs, and platelets help clotting. The other rows swap these roles incorrectly.

8. (D) The human heart has four chambers — two upper atria and two lower ventricles.

9. (A) The two upper chambers of the heart are the atria; the two lower chambers are the ventricles.

10. (D) The "lub-dub" heartbeat sound is made by the heart's valves snapping shut as blood is pumped. It is not the lungs, the wrist pulse, or capillary flow.

11. (D) Keeping oxygen-rich and oxygen-poor blood on separate sides lets the heart deliver fully oxygenated blood efficiently to the body rather than a diluted mix. The other options are not real reasons.

12. (B) The heart is a muscular organ that contracts rhythmically to pump blood; it is not made of bone, cartilage or fat.

13. (D) By definition an artery carries blood AWAY from the heart. A vein carries it back; a capillary is for exchange; a ureter belongs to the urinary system.

14. (B) Arteries carry blood pumped straight from the heart at high pressure, so they need thick, elastic, muscular walls to withstand it. Valves and slow flow describe veins, not arteries.

15. (A) Veins have valves that stop blood from flowing backward — a feature arteries lack. Thick muscular walls and high pressure belong to arteries; one-cell-thin walls describe capillaries.

16. (C) Capillaries are where oxygen, food and wastes are exchanged between the blood and body cells, because their walls are extremely thin. Arteries, veins and heart chambers carry blood but do not do the exchange.

17. (A) Capillary walls are only one cell thick, which lets materials pass through easily — perfect for exchange. Thick or valved walls would block this.

18. (C) Blood is always red; haemoglobin makes it so. Veins only look bluish because we see them through the skin — the blood inside is not blue. This is a classic misconception.

19. (D) A vein carries blood back toward the heart and contains valves. Carrying blood away at high pressure and being one cell thick describe arteries and capillaries respectively.

20. (D) A pulse is the rhythmic expansion of an artery each time the heart pumps a surge of blood into it. It is not vein drainage, vein valves, or breathing.

21. (C) A healthy adult's resting pulse (heartbeat) rate is about 72 beats per minute. 12 and 7 are far too low; 220 is impossibly high for rest.

22. (D) One pulse = one heartbeat. At 72 beats per minute, in 5 minutes the count is $72 \times 5 = 360$.

23. (A) During hard exercise the muscles need more oxygen quickly, so the heart beats faster to deliver it. It does not need less oxygen or to stop clotting.

24. (A) In 15 s there are 18 beats; 15 s is one-quarter of a minute, so per minute = $18 \times 4 = 72$ beats.

25. (C) A normal resting pulse is about 72; 120 beats per minute at rest is clearly higher than the usual resting rate (it is the kind of rate seen after exertion). It is neither normal-at-rest nor impossible.

26. (D) Excretion is the removal of harmful waste products from the body. Taking in food, pumping blood and carrying oxygen are different processes.

27. (A) The kidneys are the two bean-shaped organs that filter wastes from the blood. The lungs handle gases; ventricles and capillaries are parts of the circulatory system.

28. (B) Each kidney is built from microscopic filtering units called nephrons. Stomata are leaf pores, platelets are blood cells, and ureters are tubes.

29. (C) The kidneys remove urea — a nitrogenous waste — from the blood and pass it out in urine. Glucose and oxygen are useful; haemoglobin stays in the RBCs.

30. (D) Urine forms in the kidney, travels down the ureter to the bladder for storage, then leaves through the urethra: kidney → ureter → bladder → urethra.

31. (A) The ureter carries urine from a kidney down to the bladder. The urethra carries urine out of the bladder; an artery and phloem are unrelated.

32. (A) Sweat is mostly water with some salts, so sweating removes water and salts — a form of excretion — besides cooling the body. The skin does not absorb oxygen or pump blood.

33. (D) When the kidneys fail, a machine cleans the blood by dialysis. Transpiration is a plant process; transfusion is adding blood; circulation is normal blood flow.

34. (A) Urea is produced in the body cells, carried in the blood, filtered out by the kidneys, and finally leaves the body as urine. The other sequences mix up plant tissues or organs.

35. (A) Urea is a waste that becomes harmful (poisonous) if it builds up, so the body must remove it. It is not a stored food, nor needed for haemoglobin or blood colour.

36. (B) A plant absorbs water and dissolved minerals from the soil mainly through its root hairs. Leaves make food, flowers are for reproduction, phloem carries food.

37. (C) Root hairs are thin outgrowths that greatly increase the surface area in contact with soil water, helping absorption. They are not green, have no muscles, and do not carry food.

38. (C) Xylem is the tissue that carries water and minerals upward from the roots to the leaves. Phloem carries food; stomata and the cambium do other jobs.

39. (A) Water moves through the xylem upward, from the roots to the leaves — a continuous one-way pipe driven by transpiration pull.

40. (B) The loss of water vapour from leaves through tiny pores (stomata) is called transpiration. Respiration, excretion and germination are different processes.

41. (D) The tiny pores on a leaf through which water vapour escapes are the stomata. Root hairs absorb water; nephrons and capillaries are animal structures.

42. (B) As water evaporates from the leaves it creates a suction — the transpiration pull — that draws more water up the stem. The other terms are circulatory, not plant, ideas.

43. (B) Heat, dryness and wind all speed up evaporation, so transpiration increases and water is pulled up faster. It does not stop, reverse, or have no effect.

44. (A) Phloem carries the food made in the leaves to all other parts of the plant. Xylem carries water; root hairs and stomata are not transport tissues for food.

45. (D) Food can move through the phloem both upward and downward, to wherever in the plant it is needed — unlike the one-way upward flow of water in xylem.

46. (D) Because phloem can transport food in more than one direction at once, sugar from a leaf can reach both a root tip (down) and a fruit (up/sideways). Xylem only moves water upward.

47. (B) Removing a ring of bark cuts the phloem, so food made in the leaves can no longer travel down to the part below the ring, which then starves. The xylem (deeper, carrying water up) is left intact.

48. (A) The correct pairing is: xylem carries water and minerals, phloem carries food. The other options swap or confuse the cargoes.

49. (B) Root hairs absorb water and minerals from the soil, so a plant with damaged root hairs will most directly struggle to absorb water and minerals. Food-making, transpiration and bending toward light are not the direct effect.

50. (A) Xylem moving water up is like a one-way delivery pipe, and there is no plant 'heart' pumping food around — phloem does it by other means. The options likening xylem to veins/arteries or phloem to kidneys are wrong.

51. (B) Fully correct: arteries carry blood away from the heart (usually oxygenated) and veins carry it back (usually deoxygenated). The other statements reverse the directions or the valves/pressure facts.

52. (D) Veins usually carry deoxygenated blood, but there is an exception — the vein returning blood from the lungs to the heart carries oxygen-rich blood. So 'veins only carry dirty blood' is not fully true.

53. (D) Compared with an artery, a vein has thinner walls, carries blood at lower pressure, and has valves. Thick walls, high pressure and carrying blood away describe arteries; one-cell-thin walls describe capillaries.

54. (C) An Amoeba is a single cell, so materials can be exchanged directly across its surface by diffusion — no heart, vessels or xylem are needed. It still needs oxygen and food; it has no blood.

55. (B) In a large multicellular body the inner cells are too far from the surface to get oxygen and food by simple diffusion alone, so a transport system is needed to carry supplies to them.

56. (B) Correct matching: the ureter carries urine, the xylem carries water, and the phloem carries food. The other rows mismatch the tubes with the wrong cargo.

57. (C) Both the kidneys (filtering and excreting water in urine) and transpiration (loss of water vapour from leaves) involve the movement of water. Bone, haemoglobin and platelets are unrelated here.

58. (B) Blood returning to the heart from the body carries less oxygen, because it has given oxygen to the body cells along the way. It is still red, and it has not become urine.

59. (C) On a hot afternoon the leaves can lose water by transpiration faster than the roots supply it, so the plant wilts even in moist soil. Xylem does not turn into phloem, and the soil water is not poisonous.

60. (C) Correct summary: in animals the heart pumps blood through vessels, while in plants xylem carries water up and phloem carries food. The other options swap the heart, xylem and vessel roles.

Solutions complete: 60 / 60.